

# MINI ASSESSMENT - LAB REPORT

**Name:** Mangali Bhaskar    **Department:** CSE

**Register No:** 99230040657    **Date:** 11-09-2026

## 1. AIM

To develop an interactive Python chatbot that can respond to common user queries by identifying keywords and selecting suitable responses. The chatbot uses input preprocessing, conditional decision-making, and a continuous conversation loop.

## 2. PROCEDURE

- Create a Python function to control the chatbot conversation.
- Use a while True loop so the chatbot can continuously accept messages.
- Read the user's message using input() and clean it with strip() and lower().
- Check for exit commands such as 'bye', 'exit', or 'quit' and terminate the conversation.
- Identify common chatbot intents such as greetings, asking for available services, asking for the bot's identity, and requesting study help.
- Return an appropriate response for each recognized intent using if-elif conditions.
- Use an else branch to provide a fallback response when the chatbot cannot recognize the request.

## 3. FLOWCHART



## 4. PYTHON CODE

```
def chatbot():
    print("Bot: Hello! I am your study assistant.")
    print("Bot: Type 'help' to see what I can do.")
    print("Bot: Type 'bye' or 'exit' to end the chat.\n")

    while True:
        user_input = input("You: ").strip().lower()

        if user_input in ["bye", "exit", "quit"]:
            print("Bot: Goodbye! Keep learning!")
            break

        elif user_input in ["hello", "hi", "hey"]:
            print("Bot: Hello! What would you like to know?")

        elif "help" in user_input or "services" in user_input:
            print("Bot: I can greet you, explain my purpose, "
                  "and provide basic study-related help.")

        elif "your name" in user_input or "who are you" in user_input:
            print("Bot: I am a Python-based educational chatbot.")

        elif "python" in user_input:
            print("Bot: Python is a high-level programming language "
                  "used for software development, automation, and AI.")

        elif "study" in user_input or "learn" in user_input:
            print("Bot: You can learn step by step by practicing concepts "
                  "and solving small problems.")

        else:
            print("Bot: I don't have a response for that yet. "
                  "Try asking about Python, study help, or my services.")

if __name__ == "__main__":
    chatbot()
```

## 5. RESULT

The chatbot successfully accepts user messages and produces different responses according to recognized keywords. For example, the bot responds to greetings, explains its available services, identifies itself, provides basic Python/study information, and ends the conversation when the user enters an exit command.

## 6. CONCLUSION

A simple interactive chatbot was successfully developed using Python. The project demonstrates how input preprocessing, conditional statements, keyword matching, loops, and fallback handling can be combined to create a basic conversational interface. The design can be extended later with more intents, a larger knowledge base, or natural-language processing techniques.